A Block-Encoded Carleman–Lattice Boltzmann Solver for the Convecting Taylor–Green Vortex

Concept Proposal and Preliminary Feasibility Results for the Airbus Global Quantum + AI Challenge 2026

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Abstract

We propose an end-to-end fault-tolerant quantum algorithm for the two-dimensional convecting Taylor–Green vortex based on D2Q9 lattice Boltzmann dynamics, order-two Carleman linearization, structured block encoding, reversible streaming, amplitude amplification, and direct observable estimation. The objective is not to infer advantage from qubit count, but to demonstrate—or falsify—lower time-to-solution or memory at matched $(\mathrm{Re}, t, \epsilon)$ after all preparation, timestepping, routing, fault-tolerance, and measurement costs. Eleven executed notebooks establish a reproducible starting point: an actual local eight-qubit dilation has conditional fidelity above $1 - 10^{-12}$ through ten applications; exact amplification raises one-step success from 0.632% to 99.66% using 19 block calls; factorization reduces stored collision coefficients by 12.37×; and physics sweeps cover Re = 10–5000. The same evidence exposes the decisive risks: unamplified ten-step success is 1.01×10^{-22}, ring-routed streaming reaches depth 987,743 at $N = 128$, and a spectral baseline is faster and roughly seven orders more accurate than the current Python BGK solver. The proposed programme therefore focuses on the missing position-aware Q oracle, integrated collision–streaming circuit, direct energy/mode extraction, and fixed-accuracy crossover test. No quantum advantage is claimed from the preliminary results; the deliverable is a working solver and an auditable advantage or no-go determination. **Keywords:** quantum computational fluid dynamics, lattice Boltzmann method, Carleman linearization, block encoding, Taylor–Green vortex, fault-tolerant quantum computing.

1 Challenge Response and Proposed Outcome

Airbus seeks a working solver for the convecting Taylor–Green vortex, scaling with Reynolds number, and comparison against state-of-the-art classical methods [1]. Our response is a fault-tolerant LBM concept with a strict evidence boundary:

- **Measured**: analytic benchmark, D2Q9 solver, local Carleman dilation, amplification, raw-f sampling, routed streaming circuits, and spectral comparison.

Table 1: Traceability to the requested outcomes.

Requested outcome	Proposed evidence at completion
Working solver	Integrated raw-f PREPARE–collision–streaming–measurement circuit, verified against the exact vortex.
Scaling	Time, memory, and error versus Re = 10–5000, N = 32–2048, $t \leq 1$, and target precision.
Comparison	Optimized spectral and high-order finite-volume CPU/GPU baselines at matched error, with confidence intervals.
Advantage	End-to-end crossover, or an explicit no-go boundary, under three fault-tolerant hardware scenarios.

- **Proposed**: scalable analytic state preparation, coherent R/Q collision, global position-controlled streaming, direct observable estimation, and integrated multi-step execution.

- **Gate**: advantage is accepted only if the complete quantum ledger beats an optimized classical solver at the same physical error and observable.

The convecting vortex is the required verification case, not evidence of a production high-Mach capability. D2Q9 BGK is used here in its weakly compressible regime; extension to compressible lattices and extreme flight conditions is a follow-on only after the solver passes the present gates.

2 Benchmark and Numerical Model

For velocity $\boldsymbol{u} = (u, v)$, density ρ, pressure p, and viscosity ν, the incompressible equations are

$$\nabla \cdot \boldsymbol{u} = 0, \tag{1}$$

$$\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\rho^{-1}\nabla p + \nu\nabla^2\boldsymbol{u}. \tag{2}$$

With $\xi = (x - U_c t)/L_c$, $\eta = (y - V_c t)/L_c$, and $A(t) = \exp(-2\nu t/L_c^2)$, the exact convecting vortex is

$$u = U_c + V_0 \sin\xi \cos\eta\, A(t), \tag{3}$$
$$v = V_c - V_0 \cos\xi \sin\eta\, A(t), \tag{4}$$
$$p = p_0 + \frac{\rho V_0^2}{4}[\cos(2\xi) + \cos(2\eta)]A(t)^2. \tag{5}$$

We set $L_{\text{box}} = 2\pi$, $L_c = 1$, $V_0 = U_c = \rho = 1$, $V_c = p_0 = 0$, and $\nu = 1/\text{Re}$. Separating box and vortex scales preserves periodicity.

The D2Q9 BGK update is

$$\rho = \sum_i f_i, \qquad \rho\boldsymbol{u} = \sum_i f_i \boldsymbol{e}_i, \tag{6}$$
$$f_i^\star = (1-\omega) f_i + \omega f_i^{\text{eq}}, \tag{7}$$
$$f_i(\boldsymbol{x} + \boldsymbol{e}_i, t + \Delta t) = f_i^\star(\boldsymbol{x}, t), \tag{8}$$

where f_i^{eq} is the standard second-order equilibrium [2]. Tests cover $\text{Re} = 10, 100, 400, 1000, 2000, 5000$, Mach 0.1–0.0125, and periodic grids. Velocity, pressure, kinetic energy, momentum drift, Fourier phase/amplitude, positivity, and convergence order are the fixed validation observables.

3 Proposed Quantum Solver

Near reference density ρ_0, write $\boldsymbol{f}^{\text{eq}} \simeq L\boldsymbol{f} + Q : (\boldsymbol{f} \otimes \boldsymbol{f})$. With $R = (1-\omega)I + \omega L$ and $\boldsymbol{z} = [\boldsymbol{f}; \boldsymbol{f} \otimes \boldsymbol{f}] \in \mathbb{R}^{90}$,

$$\boldsymbol{z}^\star = M\boldsymbol{z}, \qquad M = \begin{bmatrix} R & \omega Q \\ 0 & R \otimes R \end{bmatrix}. \tag{9}$$

For $\alpha \geq \|M\|_2$, a unitary dilation block-encodes M/α. The physical populations use raw signed amplitudes, not the legacy $\sqrt{f}$ encoding; raw-f reproduced the local collision to 1.43×10^{-16}, whereas $\sqrt{f}$ incurred relative error 4.91.

The algorithm prepares the analytic initial vortex using reversible arithmetic, applies collision to all sites in superposition, streams with controlled modular shifts, amplifies the valid block, and estimates only requested observables. The central technical contribution will be coherent loading of the 496 nonzero Q coefficients and composition with reusable R and $R \otimes R$ operations, avoiding generic dense synthesis.

4 Preliminary Feasibility Evidence

4.1 Circuit success and extraction

A genuine Qiskit circuit applies the complete *local* 90-dimensional dilation with a fresh signal ancilla per application. Conditional fidelity is unity to numerical precision, but raw success compounds catastrophically (Figure 2). Exact amplitude amplification peaks after nine iterations (19 block calls) at $p_{\text{succ}} = 0.9966$.

Raw-f samples retained the state norm. From 10^3 to 10^6 shots, velocity RMSE fell 0.136 to 0.00342, kinetic-energy absolute error 9.33×10^{-3} to 1.19×10^{-6}, and $(1, 1)$ Fourier-mode error 0.936 to 0.00655. These tomography-scale results validate the encoding but do not establish efficient extraction; the proposal replaces them with direct amplitude/phase estimation.

4.2 Structure and routed streaming

The flattened 90×90 collision matrix is 88.1% nonzero. Computing the lower lift as RF_2R^T and storing R, Q reduces coefficients from 7,138 to 577 (12.37×), with maximum multiplication error 1.44×10^{-15}. A reusable five-qubit R dilation compiles to depth 825 and 423 CX gates, versus depth 107,126 and 87,216 CX for the complete dense reference. This supports structured construction, but is not yet a full PREPARE/SELECT implementation.

All modular shifts were decomposed to native one-qubit/CX gates and routed. The constant high-level depth of 29 expands rapidly after decomposition (Figure 3); topology must therefore enter every resource claim.

4.3 Physics and classical benchmark

The production notebook runs $t = 1$ for every requested Reynolds number and Mach value at $N = 32$ with five timing repetitions, plus $N = 16, 32, 64$ convergence tests. Velocity convergence orders are 1.67–1.91; no final population is negative. Best velocity L_2 error across Re is 0.00368–0.00432 and Fourier-mode error 0.00661–0.00709. Relative pressure error remains 0.312–0.314, so pressure fidelity is a specific optimization target.

A dealiased Fourier-vorticity RK4 solver provides an independent classical comparator. At $N = 64, t = 1$ it runs in 0.636–0.665 s versus 4.31–4.80 s for the Python BGK solver, while attaining $\simeq 10^{-10}$ relative velocity error rather than 3.3–4.6 $\times 10^{-3}$ (Figure 4). This is a deliberately strong baseline, not a straw man.

5 Advantage Hypothesis and Scaling Test

For $D = 9N^2$ populations, the quantum state uses $O(\log D)$ address qubits, but this is only a memory opportunity if state preparation, updates, and extraction preserve it. The complete runtime ledger is

$$T_Q = T_{\text{prep}} + n_t(T_{R/Q} + T_{\text{AA}} + T_{\text{stream}}) + T_{\text{extract}}. \tag{10}$$

The classical ledger includes initialization, solver, communication, and the same observable extraction. The proposal tests three linked hypotheses:

1. analytic TGV preparation scales polylogarithmically in N, while amplitude estimation targets $O(1/\epsilon)$ oracle calls rather than the $O(1/\epsilon^2)$ scaling of direct sampling;
2. factorized R/Q collision and modular streaming avoid dense $O(D^2)$ synthesis and retain a sublinear-in-D per-step logical cost;
3. amplification, timestep count, and error correction do not erase those savings at fixed physical error.

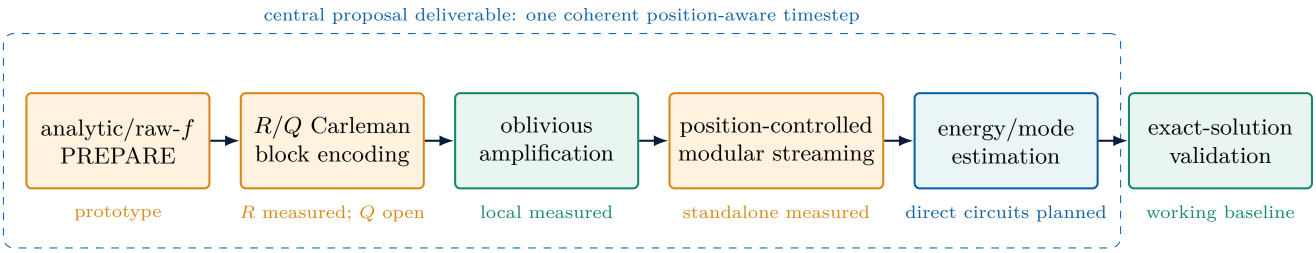

Figure 1: End-to-end solver architecture and evidence status. Orange blocks are prototypes requiring integration; green blocks are executed; blue is planned observable extraction to be implemented.

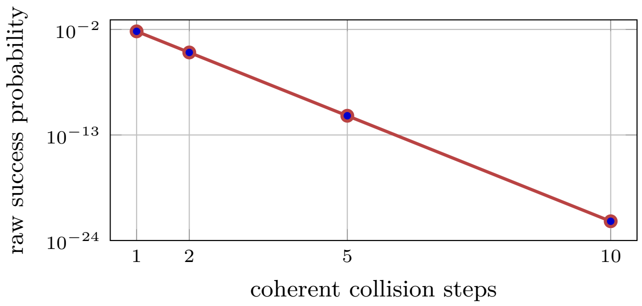

Figure 2: Measured local-circuit post-selection decay. This is the primary amplification and normalization risk.

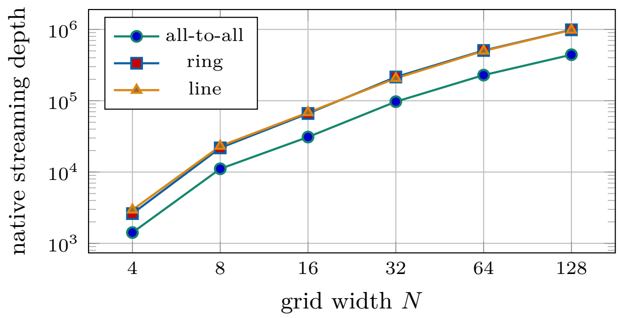

Figure 3: Measured routed D2Q9 streaming compilation.

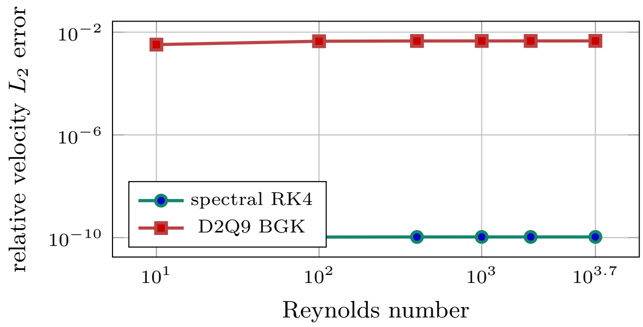

Figure 4: Measured $N = 64, t = 1$ accuracy against the exact solution.

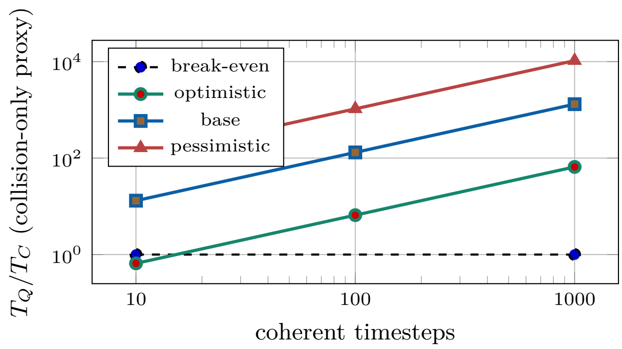

Figure 5: Preliminary no-go boundary. A final claim must add every term in Equation (10) and compare at matched error.

The hypotheses are rejected if any required oracle hides $O(N^2)$ data loading, if coherent error or success normalization grows prohibitively with n_t, or if the end-to-end ratios fail

$$\mathcal{A}_T = T_Q/T_C < 1, \qquad \mathcal{A}_M = M_Q/M_C < 1. \tag{11}$$

The present collision-only fault-tolerant sensitivity study is a lower bound, because it omits preparation, global streaming, extraction, and inter-block routing. Median quantum/classical runtime ratios across Reynolds numbers are shown in Figure 5. Only the optimistic ten-step proxy is below one; there is currently no defensible advantage.

6 Work Plan, Deliverables, and Resources

6.1 Technical work packages

WP1—Physics and baselines.
Complete $N = 32$–2048, $t = 1$ sweeps with at least five timed repetitions; add pressure, positivity, momentum, phase, mode, and order metrics; implement optimized CPU/GPU spectral and high-order finite-volume comparators.

WP2—Integrated quantum timestep.
Build analytic raw-f preparation, coherent Q loading, factorized PREPARE/SELECT or QSVT collision, and position-aware streaming. Execute one through ten complete steps at small N.

WP3—Success and observables.
Optimize α; test fixed-point and oblivious amplification across timesteps; implement energy and Fourier amplitude/phase estimators with explicit precision scaling.

WP4—Compilation and reliability.
Sweep topology through $N = 2048$; report logical/native gates, Clifford+T count, T-depth, ancillas, routing, and approximation error. Apply noise

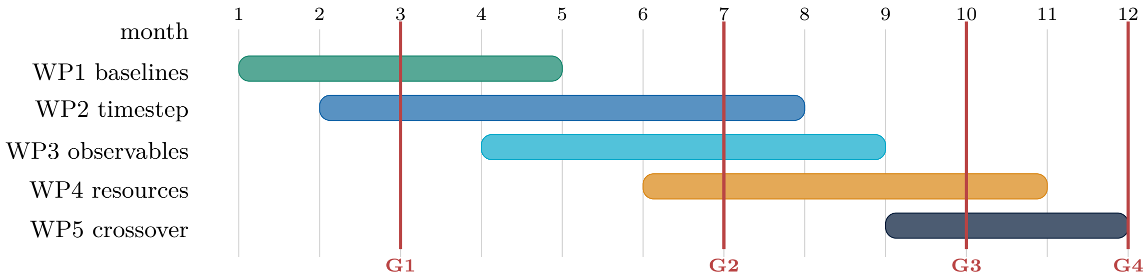

Figure 6: Twelve-month execution plan. Decision gates: G1 validated global timestep; G2 stable multi-step observables; G3 complete compiled ledger; G4 matched-error advantage or no-go decision.

after transpilation to the complete small circuit and perform a funded QPU pilot.

WP5—Crossover decision.
Combine every cost under optimistic, base, and pessimistic surface-code assumptions and compare with both classical solvers at identical $(\mathrm{Re}, t, N, \epsilon)$.

6.2 Six auditable deliverables

1. Reproducible exact-solution and classical baseline suite with hardware metadata, warm-ups, confidence intervals, and peak memory.
2. End-to-end global circuit for raw-f preparation, collision, streaming, amplification, and measurement for 1–10 steps.
3. Observable package for normalization, energy, pressure proxies, and selected Fourier modes with shot/amplitude-estimation scaling.
4. Factorized collision and streaming compiler with ancilla, topology, Clifford+T, T-depth, and approximation-error reports.
5. Stability/noise atlas over Carleman orders 1–3, Re, Mach, resolution, and 50–1000 coherent steps.
6. Fixed-accuracy crossover report and plots, including an explicit no-go result if the acceptance criteria are not met.

Required resources are an optimized multicore/GPU node for the $N = 128$–2048 baseline matrix, a circuit simulator for small integrated instances, a compiler/resource estimator with configurable topology and surface-code assumptions, and limited QPU credits for a small hardware validation. QPU data is demonstrative only; the advantage claim is based on fault-tolerant compiled resources and matched classical measurements.

7 Validation Matrix and Risk Control

All comparisons use the same domain, initial condition, t, Reynolds number, observable, and error tolerance. Five or more repetitions follow one warm-up; reports include median, range, 95% bootstrap interval, peak memory, software versions, processor/GPU, compiler seed, and circuit topology.

Table 2: Final decision gates.

Gate	Pass criterion	If failed
G1	Global $N = 4$, 1–10 step circuit: fidelity ≥ 0.99 and observables within declared error budgets	Raise Carleman order or reject the encoding/oracle.
G2	Orders 1–3 remain bounded and physical for 50–1000 steps	Adapt normalization/order or delimit the valid regime.
G3	Complete routed FT circuit meets failure target; every ledger term bounded	Redesign Q/streaming or declare resource no-go.
G4	$\mathcal{A}_T < 1$ or $\mathcal{A}_M < 1$ at matched error, with uncertainty below one	Publish crossover boundary or no-go; make no advantage claim.

Table 3: Principal risks and mitigation.

Risk	Evidence	Mitigation
Success collapse	$p_{10} = 1.01 \times 10^{-22}$ raw	Optimize α; fixed-point AA; normalize per step; account for every call.
Oracle cost	Q loading remains open	Exploit 35 coefficient magnitudes and factorized R/Q structure; compare PREPARE/SELECT and QSVT.
Routing growth	Ring depth 987,743 at $N = 128$	Arithmetic shift networks, layout co-design, and topology sweep.
Physics bias	Pressure error $\simeq 31\%$	Mach/refinement study, higher-order lift, and pressure-specific acceptance threshold.
Strong classical baseline	Spectral method already wins	Optimize both baselines; claim only a fixed-accuracy measured crossover.

8 Reproducibility and Claim Boundary

Eleven notebooks execute from clean kernels with no stored errors, and the repository test suite reports 11 passing tests. Machine-readable results are stored in `results/deliverables`. Source code, executed notebooks, and reproduction instructions are available at https://github.com/SyedT1/quantum-aero-solver. The prepared Open Quantum SDK job was not submitted because the account had zero credits; credentials are not stored in the repository. Hardware execution remains a WP4 task.

The current evidence validates local mathematical and circuit components, not the complete quantum solver. It does not demonstrate quantum advantage. The proposal is intentionally falsifiable: success means a working end-to-end solver plus a matched-error time or memory ratio below one; otherwise the project delivers a documented no-go boundary identifying which cost prevents advantage.

9 Conclusion

This concept directly targets the Airbus benchmark with a structured FTQC solver, a working exact-solution validation stack, measured preliminary circuits, a state-of-the-art classical comparator, and explicit scaling gates. The negative preliminary results strengthen the proposal by locating the critical work: coherent Q loading, global composition, stable amplification, direct observables, and complete fault-tolerant accounting. The twelve-month programme produces either defensible quantum advantage at fixed fidelity or an equally auditable no-go result, without substituting qubit-count arguments for time-to-solution evidence.

References

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